

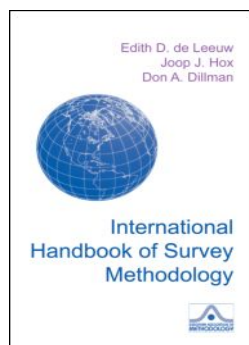
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Chapter 15

IVR: Interactive Voice Response

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15.1 INTRODUCTION

Over the past 25 years, western culture has experienced a dramatic shift in the conduct of daily functional transactions, replacing traditional interpersonal contact at places like banks, stores, and post offices with electronic self-administration such as ATMs, self check-out, and automated postal centers. Online and automated telephone systems are now the norm rather than the exception for transactions such as catalog purchases, airline reservations, bill payments, and banking. Another powerful example of this shift is the increasing popularity and usage of electronic self-administered surveys such as Web-based data collection and telephone computer-automated data collection (also known as IVR, or Interactive Voice Response). These new methods of data collection offer the power and complexity of computerization combined with the privacy of self-administration. Web and IVR technologies have become increasingly popular for studying populations that have easy access to the technology and that have a high level of willingness to interact directly with computers.

15.2 WHAT IS IVR, AND HOW DOES IT FIT INTO THE FAMILY OF DATA COLLECTION METHODOLOGIES?

15.2.1 What is IVR?

IVR is a data collection technology in which the computer plays a recording of the question to the respondent over the telephone, and the respondent indicates the response by pressing the appropriate keys on his or her touch-tone (global) telephone keypad. IVR is also referred to as T-ACASI (Telephone-Audio Computer-Assisted Self-Interviewing), TDE (Touch-tone Data Entry), and VRE (Voice Recognition Entry).

IVR is created by programming the computer to play prerecorded prompts to the respondent based on the respondent's answers. A typical survey begins with a greeting to the caller, asking for an entry to prove identity such as

a personal identification number (PIN) or password. The IVR system accesses the computer-stored data and verifies the caller's identity and password before any additional information is spoken. Only callers who are validated are permitted to continue. The survey items are then read to the caller.

IVR technology is widely used for sales and marketing, market research, customer satisfaction, employee assessment, and personnel selection, among other purposes. IVR is ideally used when the data desired are numeric or can easily be linked to a numeric code, such as "press 1 if yes, press 2 if no" (though open-ended data can also be collected via IVR), and is especially appropriate for surveys that are short and repetitive (Weeks, 1992).

IVR research can be executed in a number of different ways. The main distinctions are between *inbound* IVR studies, in which the sample member dials into an IVR system at his or her convenience, and *outbound* IVR studies, in which an interviewer dials out to the sample member to recruit respondents to participate and then transfers them directly into the IVR system. Some examples of these methods include:

- *Inbound mail recruit*: The U.S. Bureau of Labor Statistics uses IVR (referred to as TDE) for the Current Employment Statistics (CES) survey. Respondents receive a monthly advance postcard or fax reminding them to call in to the TDE system to provide payroll data. Nonrespondents receive telephone or fax prompts. (Phipps & Tupek, 1991).
- *Inbound recruit and transfer*: Using IVR (referred to as T-ACASI), customers calling in to the National Energy Information Center are transferred to an IVR system to get their feedback on the Energy Information Administration's products and services (Weir, Laurence, & Blessing, 2000).
- *Outbound CATI recruit*: Recent customers at a bank are recruited by live interviewers to provide customer feedback and are transferred to an IVR survey to provide their responses (Tourangeau, Rips, & Rasinski, 2000).
- *Inbound customer receipt recruit*: A national discount retail chain prints a toll-free number on a random sample of customer receipts to solicit feedback on their retail experience. Customers are asked to dial in within 24 hours of their store visit to provide feedback on their visit.
- *On-site inbound recruit*: Outreach workers visiting sites serving homeless adults offer them a cellular phone to take a survey about substance abuse treatment needs (Alemagno, Cochran, Feucht, Stephens, & Wolfe, 1996).

IVR has a number of strengths and limitations, which are discussed next.

15.2.1.1 Strengths

The key advantage to conducting research using IVR technology is that no human intervention is needed during data collection. Thus, the costs of data collection tend to be lower for IVR studies than for comparable studies employing live interviewers. Furthermore, inbound IVR data can be collected at any time of day or night, at the discretion of the survey respondent. Outbound

IVR studies are generally conducted during regular hours, because a live interviewer is typically present to transfer the respondent into the IVR survey.

IVR is best suited for surveys consisting primarily of closed-ended questions with a small set of response choices. IVR is an extremely efficient methodology for cross-cultural studies being conducted in multiple languages, since it is relatively inexpensive to translate and program a survey into additional languages.

15.2.1.2 Limitations

Because IVR technology relies on the usage of a touch-tone telephone in order to transmit the information, IVR is not a viable method of data collection among populations lacking a high prevalence of touch-tone telephones.

The company hosting the IVR system must have the capacity to accept a large volume of incoming calls. See this book's website, Chapter 15, Exhibit A, for a list of key features and components of a successful IVR system.

Respondent burden is an important issue to consider with IVR surveys. Without an interviewer to motivate the respondent to carry through to the end of the survey, long IVR surveys can suffer from high break-off rates. IVR should thus be limited in its usage to shorter questionnaires. IVR is less suitable for questions with more than five to seven response categories and many open-ended items. The IVR researcher must be very cautious in designing an IVR questionnaire to ensure that the respondent has a clear understanding of how to navigate the system, including how to have a question repeated, and how to skip an item.

15.2.2 How Does IVR Fit into the Family of Data Collection Methodologies?

IVR is somewhat of a hybrid data collection method. It is comparable to a CATI (computer-assisted telephone interviewing) survey in that it is administered by telephone, but is also similar to Web or other self-administered surveys, because no live interviewer is present to collect the data.

15.2.2.1 IVR vs. CATI

Like computer-assisted telephone interviewing (CATI), IVR is administered by telephone and requires the respondent to rely on an aural presentation of the survey questions. It also uses programming technology similar to CATI to allow for skip patterns and the use of sample frame information to feed into the way questions are asked. Furthermore, IVR data collection can produce an instant data file, as does CATI, upon completion of the interview, unlike paper surveys that need to be scanned or manually data entered upon receipt. Finally, like CATI, a responder to an IVR survey does not know how far along he or she is in the survey instrument unless the recorded voice gives some indication of that progress. This is a key difference from a paper survey, in which the respondent can physically see how many questions are left. If the IVR respondent senses that the survey is long and has no indication of how many questions are left, he or she may be more likely to break off the interview. The same could occur in a CATI interview, though the interviewer can easily

provide an indication of progress to the respondent.

In spite of these similarities to CATI interviewing, a key difference between CATI and IVR interviewing is the presence of a live interviewer. Live CATI interviewing can be subject to interviewer discrepancies in how interviewers read the questions, such as speed, tone, and personality that may come through as the questions are being read. Though CATI interviewers are trained to read every question exactly as worded, they sometimes do not, which may affect the quality of data collected. IVR interviewing, in contrast, uses a single recorded voice to deliver exactly the same reading of the questions every time. In addition to the potential for interviewer variability, there has been much research conducted to examine the impact of an interviewer's presence on survey statistics. The removal of the live interviewer in the IVR mode can potentially lead to more accurate reporting, particularly for items of a sensitive nature. In these respects, IVR interviewing is vastly different from CATI.

To summarize, the key advantages of IVR over CATI include: (a) for inbound IVR studies, respondents can complete the interview at their convenience, at any time of day or night; (b) IVR provides significant cost savings by eliminating the need for an interviewing workforce; (c) IVR provides consistent delivery of survey questions without any interviewer variability or unintended influence; and (d) IVR may provide more honest reporting on sensitive items.

Important disadvantages of IVR relative to CATI are: (a) for inbound IVR studies a reliance on the respondent to initiate the call, resulting in diminished response rates; (b) a lack of interviewer presence to motivate respondents to complete the interview, resulting in higher break-off rates and item nonresponse rates; (c) IVR is not suitable for lengthy or complex surveys; and (d) some respondents do not have access to a touch-tone phone.

15.2.2.2 IVR vs. paper-and-pencil

The main similarity between IVR data collection and traditional paper-and-pencil mail surveying (PAPI) is the absence of a live interviewer. Both of these modes are apt to produce less socially desirable responses than modes employing live interviewers. Inbound IVR studies are similar to paper mail surveys in that the respondents can complete the survey at any time of day at their convenience. Outbound recruit-and-switch IVR surveys, however, require the sample member to be available at the time of the interviewer's call.

In contrast, the key difference between IVR and PAPI is the method of delivery of the questions. The aural format of IVR requires a much different set of questionnaire design techniques than the visual format of paper surveys, and does not require respondents to be able to read or see. IVR questions must generally be shorter, with fewer response categories.

The main advantage of IVR over PAPI is that data are collected electronically and can be immediately processed with no delay from mail time or scan time.

Some disadvantages of IVR relative to PAPI are: (a) IVR is not suitable for lengthy surveys; (b) IVR may be more subject to context effects, since there is no visual presentation of survey items; (c) some respondents do not have access to a touch-tone phone; and (d) the respondent generally must complete

the IVR survey all in one sitting, whereas the paper survey can be done over time.

15.2.2.3 IVR versus Web

Both IVR and Web-based data collection allow computer-programmed delivery of questions without the presence of an interviewer. Skip-pattern logic can be employed and both methods are appropriate for asking sensitive items. They are also similar in that they are both relatively new data collection technologies that are still being explored and experimented with in order to determine the most effective way to use the technologies.

The key difference between IVR and Web-based data collection, again, is the aural versus visual format. IVR typically requires shorter question wording and shorter response categories than Web surveys.

Some advantages of IVR over Web-based data collection include: (a) with outbound interviewer recruiting, respondents can be immediately transferred into the IVR system, with no delay (as opposed to waiting for the respondent to log onto the Web survey); and (b) touch-tone telephone penetration is typically higher than computer penetration among households.

Key disadvantages of IVR relative to Web are: (a) IVR is not suitable for lengthy or complex surveys; (b) IVR may be more prone to context effects, since there is no visual presentation of survey items; and (c) some respondents do not have access to a touch-tone phone.

15.3 IVR AND SOURCES OF SURVEY ERROR

Surveys are subject to various types of errors (Groves, 1989). *Observational* errors are measurement errors that arise when survey responses differ from the true statistic, stemming from interviewer bias, bias in question wording, or bias related to the mode of data collection. *Nonobservational* errors arise because part of the population failed to be measured, for reasons such as being excluded from the sample frame, not having the technology to participate in the survey, or choosing not to respond (cf. de Leeuw, Hox, and Dillman, Chapter 1).

This section lays out the potential for these types of errors in IVR surveys as compared to other modes of data collection in order to aid the IVR researcher in making careful decisions about when and how to best use IVR as a data collection method.

15.3.1 Measurement Error

Measurement error can occur from a variety of sources, including the interviewer, the questionnaire, the respondent, and the mode of data collection.

15.3.1.1 Interviewer bias

Although IVR interviewing does not use live interviewers, it can still be subject to bias if the choice of voice somehow influences response. The voice can be the key factor that keeps the respondent's interest for a sufficiently long time to complete the survey. There have been a number of studies to experiment with

the use of voice in IVR research. The prevailing hypothesis in those studies was that the gender of an automated voice in an IVR survey can lead to stereotyping and thus influence responses. Tourangeau, Couper and Steiger (2001) tested this hypothesis in a series of Web and IVR experiments, varying the gender of the “presence” of an interviewer. In the IVR setting, respondents were assigned to hear either a male voice reading the questions, a female voice, or a blend of male and female voices reading questions and response options. The survey asked questions about gender attitudes, sensitive items about drug use, and other questions known to elicit socially desirable reporting. The authors found no impact of the voices on gender attitude scales or on reporting of embarrassing behaviors.

Similarly, other studies have failed to find any effect of the choice of voice on reporting levels and response rates (see Turner, et al., 1998). In a further twist on the use of voice in IVR research, Tourangeau, et al., (2001) experimented with personalization of question wording in an IVR setting, varying whether respondents heard “Now I will ask you...” to more closely mimic an interviewer-administered survey as opposed to “Next, please answer...” as a more generic application. They found that personalization of question wording in an IVR survey resulted in significantly fewer embarrassing admissions. This suggests that even though IVR is a standardized method of delivering survey questions, the level of personalization in the way questions are recorded can influence responses.

Couper, Singer, and Tourangeau (2004) also experimented with the use of a human versus a synthetic voice in an IVR experiment and found no differences in break-off rates or in reporting of embarrassing admissions between the human recording and the synthetic voice recording, in spite of respondents clearly being able to distinguish between the human and machine-generated voice. The authors concluded that synthetic voice recordings in an IVR application are a feasible alternative to human recordings.

Recommendations: IVR research calls for a straightforward and serious recording that is easy to understand. This generally is best achieved by a human voice because it is recorded in a way to achieve a higher quality of conversation with the respondent than is true for a synthesized voice. IVR survey designers should minimize the level of personalization in the IVR script, especially for surveys covering sensitive topics. Respondents can detect age and locality in the voice of the IVR interviewer; thus, choose a voice that emulates trust and commitment with respondents. Finally, given the mixed findings in the literature, researchers should carefully consider the gender of the voice being used to read the questions, as well as whether a synthetic or a human voice should be used.

15.3.1.2 Questionnaire bias/error

According to Dillman (2007), different modes of data collection require different standards for designing questions (See also Dillman, Chapter 9). CATI favors shorter scales, scales with only the endpoints labeled, unfolding of questions, yes/no items instead of mark all that apply, and generally short

questionnaires. IVR, according to Dillman, favors even shorter wording formats than telephone, even shorter scales, and even shorter questionnaires. Exhibit B (this book's website, Chapter 15) presents some examples of how questions are often constructed differently across modes, demonstrating the complexities of administering certain items in IVR.

Recommendations: There are many factors to consider when choosing wording for an IVR script. Complete sentences should be kept brief, simple, and to the point. Parentheses, slashes, or abbreviations cannot be used in an IVR script, because these items will not be read by the voice talent. All questions will be recorded exactly as the script is written. Also, clearly and consistently indicate in the script how respondents should provide their response. Response options are typically presented in ascending order, with the content mentioned first, then the key value. For example, "If you are male, press '1,' if you are female, press '2.'" When choosing words, consider how they sound and their meaning. For example, it is difficult to hear the difference between "literate" and "illiterate" on a recording.

15.3.1.3 Respondent Bias/Error

There are several respondent-driven factors that may lead to errors in an IVR setting. First, respondents may not be comfortable or familiar with the technology. Second, the use of a synthetic voice may be more difficult to understand than natural speech and may lead to problems with comprehension and memory as the respondent tries to understand the question. Some of these errors will subside as respondents become more familiar with the technology. In a study to identify respondent problems using a touch-tone system, Phipps and Tupek (1991) asked respondents to respond both by TDE as well as by returning a paper survey form. They compared the TDE data with the survey forms and found the largest number of errors in TDE item nonresponse (82/177). An additional 18 respondents typed in too few or too many digits, and 17 slipped on the keypad and mistyped their responses. They also found that errors were reduced with experience and suggest that a panel survey may be the most appropriate for this method of data collection.

The design of some telephones may be another potential source of error. Many households have touch-tone telephones with the keys on the handset rather than the base, so the respondent needs to pull the phone away from his or her ear to press the response. Participants report higher difficulty answering the questions because of this constant moving of the handset away from their ears to press their responses (Mingay, 2000).

Recommendations: It is critical to include a practice item at the beginning of the IVR script to help respondents get accustomed to the methodology. The IVR system should allow a sufficient number of seconds for the respondent to enter their response, which accounts for the physical movement of the phone from ear to data entry. All IVR instruments should be pretested to ensure a smooth, error-free administration of the instrument.

When asking questions that are very personal to the respondent, such as income or age, it is usually best to ask these questions at the end of the survey, and to give an option to the caller who chooses not to respond. For example, “If you choose not to respond, press ‘0.’” This same type of option can also be given for open-ended questions, such as “If you would like to provide a response, press ‘1.’ If not, press ‘0.’”

Designate a standard key to be used to repeat the question and inform the respondent of this option at the beginning of the survey. Typically, this is the asterisk (*) on the keypad. For sample IVR default scenarios and suggestions for conveying instructions to the respondent, see Exhibit C, this book’s website (Chapter 15).

15.3.1.4 Bias Resulting From Mode of Administration

Respondents may be more willing to honestly respond to sensitive questions if they believe their responses will be anonymous or confidential (see also Lensvelt-Mulders, Chapter 24). A large body of studies, in fact, shows that more embarrassing (or less socially desirable) behaviors are reported at a higher rate in a self-administered survey environment than in an interviewer-administered environment (Turner, Miller, & Smith, 1996; Tourangeau & Smith, 1998; Gribble, Miller, Codey, Catania, Pollack, & Turner, 2000; Cooley, Miller, Gribble, & Turner, 2000; Currivan, Nyman, Turner, & Biener, 2004).

Table 15.1 Respondent preferences for T-ACASI versus standard telephone interview using a human interviewer

Dimension	Preferences for			ODDS RATIO (a)
	T- ACASI	Human	Indifferent	
	%	%	%	
Best at protecting your privacy	49	11	40	4.53
Best for getting honest answers	73	17	10	4.39
Best for asking about sensitive topics like sexual behavior	66	23	11	2.88
More comfortable giving your answers	44	24	31	1.82
Easier to use	30	59	11	.51
Most interesting to use	27	50	23	.54
Easiest to change answer (b)	1	61	37	.02

(a) Odds ratio for choice of T-ACASI. Indifferent responses were excluded.
(b) This question asked respondents to rate, which was the "hardest" mode in which to change answers. Authors report responses for the "easiest" mode in order to make them consistent with the coding of other dimensions in the table.
Source: Turner et al. (1996).

According to Turner, et al. (1996), respondents thought T-ACASI was better than CATI at protecting privacy, was a better environment for answering sensitive questions, made them more likely to report sexual behaviors, and was a more comfortable environment for giving answers. They also felt CATI interviewing was easier and more interesting to use, and was easier for changing answers.

With slightly less sensitive items that may still be subject to some social desirability bias, mode comparison experiments have shown mixed results, with some showing lower and perhaps more honest satisfaction ratings on IVR than CATI (Tourangeau, Steiger, & Wilson, 2002; Weir, Laurence, & Blessing, 2000). Other studies have suggested that IVR and CATI produce similar responses (Dillman, Phelps, Tortora, Swift, Kohrell, & Berck, 2002) but that IVR and CATI both produce more extreme responses than visual modes of Web and paper (see also Srinivasan & Hanway, 1999). The theory is that in a visual environment, respondents are more likely to consider the entire scale of response options than in an aural mode when factors such as primacy, recency, acquiescence bias, and social desirability may be at play.

Not all survey questions are subject to social desirability bias. For less sensitive items, IVR is considered to be an excellent alternative to the costly CATI method of data collection. The Bureau of Labor Statistics has been collecting reliable employment statistics using IVR technology since 1987. Businesses are asked to provide straightforward statistics each month in a survey that usually takes less than two minutes. Record check studies show the average error rate is less than 1.8% (Phipps & Tupek, 1990). Mingay (2000) shows several other studies that suggest that the reliability and validity of IVR as a method to collect nonsensitive information is no different from modes using a live interviewer.

Recommendations: Researchers should generally be cautious about using more than one methodology to collect survey data, especially for topics that may be sensitive in nature. IVR data can be significantly different from interviewer-collected or visually administered survey data.

15.3.2 Nonresponse Error

Nonresponse error occurs when some sample members cannot be located, refuse to participate, or are otherwise unavailable to participate in the survey during the data collection period. When nonrespondents are different from respondents, this can affect the survey statistics. Nonresponse can occur at the unit level, meaning no response is obtained from that sample member, or at the item level, meaning questions on the survey are skipped or do not have a legitimate response.

15.3.2.1 Unit nonresponse

When considering the choice of mode, one must consider respondents' willingness to participate using that method of data collection (cf. Lynn, Chapter 3). For example, sample members may be much more likely to

participate in a survey if called by a live interviewer who persuades them of the importance of participation than if handed a receipt at a store with an 800 number to dial to complete a survey. There certainly is a tradeoff with costs, because using interviewers to dial out can increase the cost of what should otherwise be a less expensive methodology than interviewer-administered CATI surveys. Exhibit D (this book's website, Chapter 15) summarizes recent research into differential response rates by mode of data collection. IVR typically produces lower response rates than CATI, but can also achieve extremely high response rates if executed carefully.

15.3.2.2 Item nonresponse

Item nonresponse in a CATI survey arises when the respondent notifies the interviewer that he or she does not have an answer to the question or would like to refuse to provide a response. In a visual environment such as Web or paper-and-pencil, the respondent may simply leave an item blank or mark off a "don't know" box if he or she chooses not to provide a legitimate response. IVR respondents, however, must become familiar with the norms of the technology in order to perform these basic functions.

In a recent study, Mingay and Kim (1998) found that in spite of reporting that they understand how to respond to an automated interview, respondents might not recall or realize how to repeat questions, change answers, or indicate they would like to skip a question. Indeed, Couper, Singer, & Tourangeau (2004) found that with no human present to prompt the respondent to provide an answer and with the possibility of technical errors (e.g., pressing an out-of-range key), missing data rates were significantly higher for the IVR conditions than for CATI (more than 2% of the answers were missing in the three IVR conditions versus about 0.5% in the CATI condition).

Recommendations: In order to minimize item nonresponse, two guiding principles are: keep the questionnaire short, and ask short questions with a short list of response categories. Additionally, the survey designer should write simple instructions into the IVR script about how to skip a question. This may be necessary to do more than once, but not too often or else respondents may be encouraged to skip items.

15.3.2.3 Break-off rates compared to other modes

IVR can provide some easy opportunities for respondents to hang up without having completed the interview. If the data collection method is outbound interviewer recruit with a transfer into the IVR system, the transfer period affords an easy opportunity for the respondent, who has already agreed to participate, to change his or her mind and hang up. This can be an especially vulnerable period if it takes more than a few seconds to complete the transfer to the automated interview. For example, Gribble, et al. (2000) found that about 18% of the sample disconnected during the transfer to IVR. Dillman, et al. (2002) portrays a typical transfer success rate of 69% when the transfer time to the IVR system is about 10 seconds long. As technology improves and this transfer time is reduced, this type of break-off rate should decline.

Tourangeau, et al. (2002) conducted an experiment in a recruit-and-switch IVR survey whereby the interviewers asked half the sample a few innocuous questions before switching them to IVR, and the other half were switched to IVR immediately upon agreeing to participate. The hypothesis was that respondents would be less likely to break off during the transfer if they had already begun the survey with the interviewer. Indeed, respondents who were asked the additional questions before the switch were significantly more likely to complete the IVR questions (68.5% versus 59.6%, $p < .001$).

Even if the transfer is successful, it is all too easy for the respondent to break off part of the way through the questionnaire if he or she finds the topic to be boring, the survey too long, or for some other distraction such as call waiting to occupy their time. Tourangeau, et al. (2002) found that most break offs occurred at two points in their survey: (a) when a lengthy explanation of a response scale was read, and (b) when a message indicated that the respondent was “now about halfway done with the interview.” These findings are consistent with earlier results that break offs are likely when respondents believe there are still many items to come (Tourangeau & Steiger, 1999).

Break offs can be computer- or respondent-generated. Nyman, Roman, & Turner (2001) recontacted respondents who had broken off the survey to find the reason for the break-off. Reasons were split between computer problems and respondents being the source of the break off. Computer break offs were mainly because the computer disconnected (51%), and call waiting (20%). Respondent break offs were mainly because the interview was too long (33%) and they did not like the questions (28%).

Break-off rates can be extremely high with a long IVR survey. Tourangeau, et al. (2002) measured a 40% break-off rate in a survey averaging 30 minutes. Cooley, et al. (2000) reported a break-off rate of 24% in a similarly long questionnaire.

Recommendations: For outbound recruiting with an immediate transfer to IVR, it is important to use an IVR system that can minimize the number of seconds to complete the transfer. It may help to engage the respondent before the transfer by having the live interviewer ask a few nonsensitive survey items. When transferring a respondent to the IVR system, it is important that the live interviewer: thanks the respondents, tells them how long the survey lasts, explains they may experience a delay in transfer (don't hang up), and confirms their willingness to complete the survey before the transfer.

15.3.3 Coverage Error

Coverage error occurs when some people are not included on the frame or are not given an opportunity to participate in the survey. Examples include lack of telephone in a CATI survey, lack of Web access on a Web survey, or lack of a touch-tone phone in an IVR survey.

15.3.3.1 Proliferation of Touch-Tone Phones, Phone Coverage

The feasibility of using IVR depends upon the number of target respondents who have access to a touch-tone telephone. According to Turner, et al. (1998), touch-tone phone ownership in the United States is over 90%, but is less common among elderly households (i.e., those without an 18- to 49-year-old) and households with people with lower levels of income and education. These rates will obviously differ across nations, and can be a source of error if the technology does not support this method.

Rotary dial registers are an option for IVR data collection, by detecting a series of clicks that are written to the data set. The cost of rotary detection depends on the penetration of rotary phones in the sample area.

15.3.3.2 Regional and Global Reach

It is important to recognize the cost of long-distance telephone calls when deciding how to manage an IVR system. For multiple incoming calls from a region, it may make sense to centralize the termination of calls to a regional hub. For example, calls from Japan may be terminated in Australia even though this is a higher rate per minute call because there may not be enough volume to warrant the expense of another system in Japan. Or given the business reasons, there may need to be several points of call termination to meet the research needs because the volume of calls to each termination point is high enough that the cost of the systems placed will offset or be less than the long distance per-minute costs.

In general, multi-nation IVR studies are often more complicated than single-nation studies. In addition to an international toll-free number (which depends on the local telephone companies working together), there is the added challenge of writing the script, translating, and recording to accommodate all respondents. And all combinations of respondents need to be considered (physically impaired or rotary phones) or based on the area, which may be technologically less advanced than other areas being sampled for the survey.

Recommendations: The rate of touch-tone phone penetration in your area should be examined before a decision is made to use this mode of data collection. For multi-nation surveys, it will be important to assess the costs of international phone calls versus adding the cost of additional IVR systems to be placed in each country.

15.4 CONCLUSIONS

15.4.1 Summary of IVR Technical Specifications

When setting up a new IVR system, the survey organization must first determine the scope of the projects likely to be using the system. Larger projects require more extensive hardware, either in-house, or supplied by a vendor. Some examples of commercial IVR software in the United States and globally include Voxeo Voice Center IVR Platform (global), Amcat IVR (U.S.,

Europe), Cyber Futuristics IVR Go4 (India), CT Developer Studio (North America), VoiceGuide IVR (Australia, U.S.), and EasyIVR (U.S.). The projected number of simultaneous callers is the main factor in determining the size of the IVR installation. It is usually wise to create a buffer in case the volume of callers goes over the planned maximum. The number and type of phone lines that will be needed will also be determined by the number of simultaneous callers. The IVR system itself is a self-contained computer server that functions much like any other computer on the network. It needs to be located in close proximity to the incoming phone lines, preferably in a climate-controlled environment.

Once the hardware has been installed and the phone lines have been tested, the system is ready for use. The IVR application must be programmed according to the company's best practices. It is usually best to develop an IVR script that contains all the questions that are required as well as the survey flow logic. Additionally, the script should contain any error messages and other instructional messages the system might need to respond with in the event that the respondent makes a mistake or needs special guidance. Remember that respondents are guided through the system by voice prompts, so the prompts need to be specific and succinct. The IVR voice prompts will need to be recorded by the chosen voice talent and the application will need to be tested for functionality prior to taking live callers. Pay close attention to the exact wording of the recorded prompts and specific pronunciations of more difficult words in the script. It is useful to have difficult words spelled phonetically on the IVR script. After it has been properly tested and the resulting data has been checked for accuracy, the survey is ready for live callers.

Since the IVR system is unattended and available 24 hours a day, it is recommended that the system be monitored for continuous operation. The survey should have a specific field period so the respondents know when they are able to call in and let their opinions be heard. Creating a successful IVR system requires a solid understanding of your project scope and attention to detail to make a robust IVR system.

15.4.2 Future of IVR

IVR is a cost-efficient way to collect data, and continues to be recognized as such in the future. As the penetration of touch-tone telephones grows within areas around the globe, IVR will become increasingly feasible as a data collection method, particularly for niche surveys in which the population of interest is known to have access to touch-tone phones.

In the United States specifically, there exists a steady stream of activity toward this method. It is particularly popular for targeting a sample with a research collection and solution product that is known to have easy access to telephones. The international survey researcher must consider the rate of touch-tone penetration and the availability of good commercial IVR systems before deciding upon IVR as a method of data collection.

GLOSSARY OF KEY CONCEPTS

Break offs. Occur when a respondent fails to complete the survey request. The break off is determined to be at the last question the respondent answered before hanging up or exiting the survey. This results in a partial interview that may or may not be counted towards the final dataset, depending on the researcher's decision.

CATI. The acronym for Computer Assisted Telephone Interviewing, in which a questionnaire to be administered by telephone interviewers is programmed into a computer system that manages the data collection and eliminates the need for interviewers to use paper and pencil to record responses.

Coverage error. Occurs when some people are systematically excluded from the sampling frame or are not given an opportunity to participate in the survey. Examples include lack of telephone in a CATI survey, lack of Web access on a Web survey, or lack of a touchtone phone in an IVR survey.

Inbound studies. Those in which the sample member dials into an IVR system at his or her convenience.

IVR. The acronym for Interactive Voice Response, which is a data collection technology in which the computer plays a recording of the question to the respondent over the telephone, and the respondent indicates the response by pressing the appropriate keys on his or her touchtone telephone keypad.

Outbound studies. Those in which an interviewer dials out to the sample member to recruit respondents to participate and then transfers them directly into the IVR system.

PAPI. The acronym for Paper and Pencil Interviewing, which can take several forms, but is typically a mailed questionnaire in which the respondent is asked to write in their responses and mail their completed questionnaire back to the data collection organization.

Respondent burden. A measure of the amount of time and effort it takes a respondent to respond to a question or a survey.

Social desirability bias. Occurs when the respondent attempts to portray himself or herself in a positive light to the interviewer.

T-ACASI. The acronym for Touchtone-Audio Computer Assisted Self-Interviewing (see IVR).

TDE. The acronym for Touchtone Data Entry (see IVR).

VRE. The acronym for Voice Recognition Entry (see IVR).